Short Notes: Structural Organization of Proteins

Overview: Protein function depends strictly on its 3D conformation, which in turn is dictated by its amino acid sequence. There are 4 levels of structure:

1. *Primary (**1°**): Sequence of amino acids.*
2. *Secondary (**2°**): Local folding (e.g., **α**-helices, **β**-sheets).*
3. *Tertiary (**3°**): 3D folding into domains.*
4. *Quaternary (**4°**): Functional assemblies of multiple subunits.*

Bonds Stabilizing Protein Structure

Protein conformation is maintained by both strong and weak bonds.

A. Covalent (Strong) Bonds

- *Peptide Bonds: Basic linkages between amino acids.*
- *Disulphide Bonds: Formed between two Cysteine residues (linking their **-SH** groups).*

B. Non-Covalent (Weak) Bonds

Weaker than covalent, but collectively crucial for stability.

- *Hydrogen Bonds: Between peptide linkages (e.g., between **N-H** of one and **C=O** of another).*
- *Electrostatic (Ionic) Bonds: Salt bridges between oppositely charged groups (e.g., $-NH_3^+$ and $-COO^-$ side chains).*
- *Hydrophobic Interactions: Non-polar side chains cluster together, moving away from water.*
- *Van der Waals Interactions: Weak attractions between closely packed atoms.*

1. Primary Structure

The linear sequence of amino acids joined by peptide bonds. It forms the backbone of the protein.

- *Has directionality: Reads from the N-terminal (free $-NH_2$) to the C-terminal (free **-COOH**).*
- *The backbone has the potential to form hydrogen bonds.*

2. Secondary Structure

Local interaction of amino acids through hydrogen bonds.

Alpha (α)-Helix

- Rod-like, coiled structure. Backbone is inner; side chains extend outward.
- Stabilized by H-bonds between the **i** and **i+4** residues.
- Key metrics:
  - **3.6** amino acids per turn.
  - Distance between adjacent amino acids = **0.15 nm**.
  - Pitch (vertical distance per turn) = **0.54 nm**.
- Usually Right-handed (clockwise rotation).

Beta (β)-Strand / Pleated Sheet

- Linear strands line up side-by-side to form a pleated sheet.
- Stabilized by H-bonds between **-NH** and **-CO** of adjacent chains.
- Parallel: Strands run in the same direction (N→C and N→C). H-bonds are zig-zag.
- Anti-Parallel: Strands run in opposite directions (N→C and C→N). H-bonds are straight.
- Distance between adjacent amino acids = **0.35 nm**.

Beta (β) Turns / Hairpin Loops

- Allow the peptide chain to change direction, making proteins globular.
- Often 2-residue hairpin loops involving 4 successive amino acids.
- Proline and Glycine are frequently found in beta turns.

3. Tertiary Structure

Complex 3D folding of the entire polypeptide chain. Involves bringing distant residues closer together.

- Globular Proteins (e.g., Myoglobin, Albumin): Water-soluble. Hydrophobic side chains are buried inside; hydrophilic groups are on the surface.
- Fibrous Proteins (e.g., Keratin, Collagen): Membrane/structural. Hydrophobic amino acids are on the surface to interact with lipid bilayers.

Structure determined via X-ray crystallography or NMR and stored in the Protein Data Bank (PDB).

4. Quaternary Structure

Found only in proteins with two or more polypeptide chains (subunits or protomers).

- Subunits can be similar or dissimilar (e.g., Hemoglobin has 2 α and 2 β chains).
- Held together by weak, non-covalent interactions (Hydrophobic, H-bonds, Ionic, Van der Waals).
- Examples: Hemoglobin, Lactate dehydrogenase.

Quick Comparison: α-Helix vs β-Strand

Feature	Alpha Helix	Beta Strand
Axial distance / residue	0.15 nm	0.35 nm
Residues per turn/sheet unit	3.6	2
Pitch (Length)	0.54 nm	0.70 nm

Polypeptides in the beta-conformation are far more extended.